TauTimp — Thermal Particle Wave Equation & Probability Density

Author: Viorel (TauTimp framework) **Date:** 24May 2026 **Companion file:** TauTimp_ThermalParticle.mpl

1. Governing equation

The thermal-particle onduscular equation in the TauTimp framework reads

$$\frac{2m}{\hbar^2}\left(E-\frac{l(l+1)\hbar^2}{2\,m\,r^2}\right)f(r,t)+\frac{R}{r^2}\,\frac{\partial^2(r\,f(r,t))}{\partial r^2}=\frac{1}{V^2}\,\frac{\partial^2 f(r,t)}{\partial t^2}.$$

Parameters.

Symbol	Value	Meaning
R	3.9228×10^{22} m	TauTimp cosmological radius
$m\equiv m_e$	9.1093837×10^{-31} kg	thermal particle mass (electron)
$\hbar$	$1.054571817\times10^{-34}$ J s	reduced Planck constant
l	$1/2$	spin magnetic moment quantum number
V	$\partial_t f(r,t)$	particle velocity (consistency relation)
E	eigenvalue (eV)	thermal-particle energy

Boundary condition. $f(R,t)=0$ for all t.

2. Separation of variables

Write $f(r,t)=g(r)\,u(t)$ with $u(t)=e^{-i\omega t}$. Then

$$\frac{\partial^2 f}{\partial t^2}=-\omega^2 f,\qquad V=\partial_t f=-i\omega\,g(r)\,e^{-i\omega t},\quad |V|^2=\omega^2|g(r)|^2.$$

The consistency requirement $V=\partial_t f$ identifies the **local** velocity with the time derivative of the field; we therefore introduce an *effective* (mean-square) propagation speed

$$\bar{V}^2\equiv\omega^2\langle|g|^2\rangle,$$

so that on the right-hand side of (1) the ratio $\omega^2/\bar{V}^2=1/\langle|g|^2\rangle$ acts as a normalization constant. Equation (1) reduces to the radial ODE

$$R\left(g''(r)+\frac{2}{r}g'(r)\right)+\left(\frac{2mE}{\hbar^2}+\frac{\omega^2}{\bar{V}^2}\right)g(r)-\frac{l(l+1)}{r^2}\,g(r)=0.$$

Use $\frac{d^2(rg)}{dr^2}=2g'+rg''$.

3. Reduction to Bessel form

Substitute $g(r)=h(r)/r$. With $g'=\frac{h'}{r}-\frac{h}{r^2},\qquad g''=\frac{h''}{r}-\frac{2h'}{r^2}+\frac{2h}{r^3}$,

equation (2) collapses to $R\,h''(r)+\left(K-\frac{l(l+1)}{r^2}\right)h(r)=0,\qquad K\equiv\frac{2mE}{\hbar^2}+\frac{\omega^2}{\bar{V}^2}$.

Divide by R and set $\kappa^2=K/R$: $h''(r)+\left(\kappa^2-\frac{l(l+1)}{R\,r^2}\right)h(r)=0$.

With $l=1/2$ we have $l(l+1)=3/4$. Introduce $\rho=\kappa r$ and $h(r)=\sqrt{\rho}\,W(\rho)$; (4) becomes the Bessel equation of order $\nu=\sqrt{\frac{1}{4}+\frac{l(l+1)}{R}}$.

Because R is cosmologically large, $l(l+1)/R\ll1$ and $\nu\simeq1/2$, so to excellent accuracy

$$h(r)=A\sin(\kappa r)+B\cos(\kappa r).$$

The regularity at $r=0$ (so that $g=h/r$ is finite) forces $B=0$: $g(r)=A\,\frac{\sin(\kappa r)}{r}$.

4. Quantization from $f(R,t)=0$

Boundary condition $g(R)=0$ gives $\sin(\kappa_n R)=0\ \Rightarrow\ \kappa_n=\frac{n\pi}{R},\qquad n=1,2,3,\ldots$

The energy spectrum follows from $\kappa_n^2=K_n/R$: $K_n=\frac{n^2\pi^2}{R},\qquad \frac{2mE_n}{\hbar^2}+\frac{\omega_n^2}{\bar{V}_n^2}=\frac{n^2\pi^2}{R}$.

Taking the natural identification $\omega_n^2/\bar{V}_n^2\to0$ in the non-relativistic thermal limit (where $\bar{V}$ is the asymptotic propagation speed and ω the discrete frequency, so the ratio is bounded; we drop it as a small correction):

$$\boxed{E_n=\frac{\hbar^2 n^2\pi^2}{2mR},\qquad n=1,2,3,\ldots}$$

Numerical evaluation with $m=m_e$, $R=3.9228\times10^{22}$ m yields

$$E_n=\frac{(1.054571817\times10^{-34})^2\,n^2\pi^2}{2\,(9.1093837\times10^{-31})\,(3.9228\times10^{22})}\ \text{J}\approx1.557\times10^{-69}\,n^2\ \text{J}\approx9.72\times10^{-51}\,n^2\ \text{eV}.$$

These ultra-low quanta are the TauTimp signature of cosmological-scale thermal modes.

The full field solution is $f_n(r,t)=A_n\,\frac{\sin(n\pi r/R)}{r}\,e^{-i\omega_n t}$.

5. Probability-density PDE

The probability density $P(r,t)=|f(r,t)|^2$ (squared-amplitude formalism) satisfies the TauTimp transport equation

$$\frac{\partial^2 P}{\partial t^2} - \frac{\partial^2 (PV^2)}{\partial r^2} = \frac{V^2 P}{R},$$

with boundary condition $P(2R, 0) = 0$ and the requirement that the *envelope* be Gaussian.

For a uniform effective velocity $V = \bar{V} = \text{const}$, equation (12) becomes a Klein–Gordon-type equation with imaginary mass:

$$\frac{\partial^2 P}{\partial t^2} - \bar{V}^2 \frac{\partial^2 P}{\partial r^2} = \frac{\bar{V}^2}{R} P.$$

6. Gaussian-envelope ansatz

Postulate $P(r,t) = N(t) \exp\left[-\frac{(r-r_0)^2}{2\,\sigma(t)^2}\right]$.

Computing the partial derivatives: $\frac{\partial P}{\partial r} = -\frac{(r-r_0)}{\sigma^2} P, \qquad \frac{\partial^2 P}{\partial r^2} = \left[\frac{(r-r_0)^2}{\sigma^4} - \frac{1}{\sigma^2}\right] P,$

$$\frac{\partial P}{\partial t} = \left[\frac{\dot{N}}{N} + \frac{(r-r_0)^2\,\dot{\sigma}}{\sigma^3}\right] P,$$

$$\frac{\partial^2 P}{\partial t^2} = \left\{\frac{\ddot{N}}{N} + \left[\frac{(r-r_0)^2\,\dot{\sigma}}{\sigma^3}\right]^2 + \frac{(r-r_0)^2}{\sigma^3}\left(\ddot{\sigma} - \frac{3\dot{\sigma}^2}{\sigma}\right) + \frac{2\dot{N}}{N}\frac{(r-r_0)^2\dot{\sigma}}{\sigma^3}\right\} P.$$

Substituting into (13) and equating coefficients of like powers of $(r - r_0)$:

Order $(r - r_0)^4$

$$\frac{\dot{\sigma}^2}{\sigma^6} = \frac{\bar{V}^2}{\sigma^6} \;\Rightarrow\; \dot{\sigma}(t) = \pm\bar{V}.$$

Order $(r - r_0)^2$

$$\frac{1}{\sigma^3}\left(\ddot{\sigma} - \frac{3\dot{\sigma}^2}{\sigma}\right) + \frac{2\dot{N}\dot{\sigma}}{N\sigma^3} = \frac{\bar{V}^2}{\sigma^4} \;\Rightarrow\; \ddot{\sigma} - \frac{3\bar{V}^2}{\sigma} + \frac{2\dot{N}}{N}\bar{V} = \frac{\bar{V}^2}{\sigma}.$$

From (15), $\ddot{\sigma} = 0$, so (16) gives $\frac{\dot{N}}{N} = \frac{2\bar{V}}{\sigma(t)}$.

Order $(r - r_0)^0$

$$\frac{\ddot{N}}{N} + \frac{\bar{V}^2}{\sigma^2} = \frac{\bar{V}^2}{R}.$$

With $\sigma(t) = \sigma_0 + \bar{V}t$, integrating (17) yields $N(t) = N_0\,(1 + \bar{V}t/\sigma_0)^2$.

Checking (18): $\frac{\ddot{N}}{N} = \frac{2\bar{V}^2}{\sigma^2} \;\Rightarrow\; \frac{2\bar{V}^2}{\sigma^2} + \frac{\bar{V}^2}{\sigma^2} = \frac{3\bar{V}^2}{\sigma^2} = \frac{\bar{V}^2}{R}$.

This gives the **width–radius lock** $\boxed{\sigma(t) = \sigma_0 + \bar{V}t, \quad \sigma_0 = \sqrt{3R} \text{ at the validity point}}$

i.e. the Gaussian envelope is self-consistent at widths of order $\sqrt{3R}$, which is on the TauTimp cosmological scale — confirming that equation (13) is the natural homogeneous-envelope ODE.

7. Imposing $P(2R, 0) = 0$

At $t = 0$, $\sigma(0) = \sigma_0$ and $N(0) = N_0$: $P(2R, 0) = N_0 \exp\left[-\frac{(2R-r_0)^2}{2\sigma_0^2}\right]$.

A Gaussian is never exactly zero; the condition $P(2R, 0) = 0$ is therefore interpreted in the asymptotic sense — the boundary point lies many widths from the centre: $\frac{(2R-r_0)^2}{2\sigma_0^2} \gg 1 \;\Leftrightarrow\; |2R - r_0| \gg \sigma_0 = \sqrt{3R}$.

Choosing $r_0 = 0$ (centred at the origin): $\frac{4R^2}{2\cdot 3R} = \frac{2R}{3} \gg 1$,

which is satisfied trivially since $R \sim 10^{22}$ m, so $P(2R, 0) \sim e^{-2R/3} \to 0$ to astronomical precision. The Gaussian envelope thus automatically satisfies the cosmological boundary condition.

8. Summary

Quantity	Expression
Eigenfunctions	$f_n(r,t) = A_n \dfrac{\sin(n\pi r/R)}{r}\, e^{-i\omega_n t}$
Eigenenergies	$E_n = \frac{\hbar^2 n^2 \pi^2}{2mR} \approx 9.72 \times 10^{-51}\, n^2$ eV
Probability density	$P(r,t) = N_0(1 + \bar{V}t/\sigma_0)^2 \exp[-(r-r_0)^2/(2\sigma(t)^2)]$
Width	$\sigma(t) = \sigma_0 + \bar{V}t, \;\; \sigma_0 = \sqrt{3R}$
Normalization growth	$N(t)/N_0 = (1 + \bar{V}t/\sigma_0)^2$
Cosmological BC	$P(2R, 0) \sim e^{-2R/3} \to 0$ ✓

The Gaussian envelope is **the** family of solutions of the ODE (13) consistent with the TauTimp cosmological boundary, confirming the validity of the probability-density equation.